6.2 Muscular System

In the body, muscular system along with the skeletal and nervous system, is responsible for the body movements.

Muscles can contract and therefore, help in moving other parts of the body. It maintains the posture and body position. There are three types of muscles namely

- **Skeletal muscle**
- **Smooth muscle**
- **Cardiac muscle**

Activity 3: Move your lower arm up and down gently. Feel the contraction and relaxation of your biceps and triceps muscles. The muscles present in the upper arm help in the contraction of front biceps muscles (become short and thick), and also relaxation of rear triceps muscles (become long and thin). You can feel the muscles on top that go stiff. When the arm is moved downwards, the front muscles relax and the rear muscles contract.

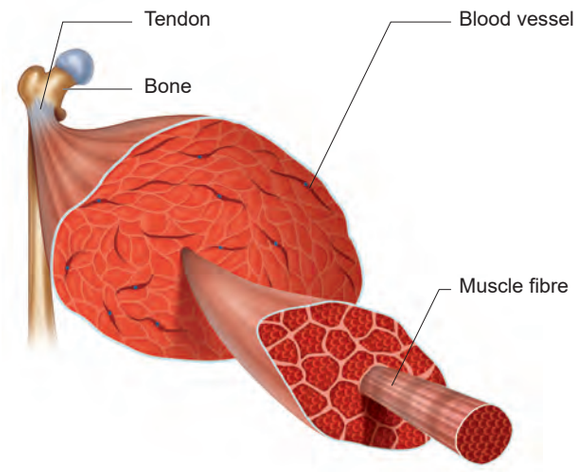
How do muscles work?

Muscles of the body can only pull and they cannot push. Two muscles are required to move a bone at a joint. When one muscle contracts, the other muscle relaxes.

For example, to move 'the lower arm up and down two type of muscles called biceps and triceps are required. When we raise our lower hand, the biceps in front become short by contraction and the triceps at the back stretch to pull up the arm. When we lower our arm, the triceps at the back contract and biceps stretch to pull the arm down.

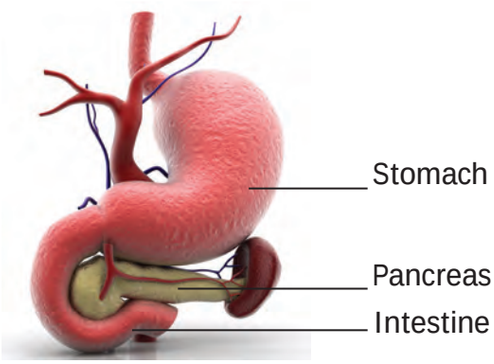
Skeletal Muscles

Skeletal muscles of our body are attached to the bones. They are called **Voluntary muscles** because they can be controlled by our will. Example: Muscles of arm.



Smooth muscles

Smooth muscles are found in the walls of the digestive tract, urinary bladder, arteries and other internal organs. They are called **'Involuntary muscles'** because they are not controlled by our will.



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